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```
MariaDB [(none)]> create database EliteVideo;  
Query OK, 1 row affected (0.015 sec)
```

Question 1:

```
MariaDB [(none)]> Use EliteVideo;  
Database changed  
MariaDB [EliteVideo]> create table MEMBERSHIP(Mem_Num int primary key, Mem_Fname  
varchar(30) NOT NULL, Mem_Lname varchar(25) NOT NULL, Mem_Street varchar(80),  
Mem_City varchar(50), Mem_State char(2), Mem_Zip char(5), Mem_Balance Decimal(8,2)  
Default 0.00);  
Query OK, 0 rows affected (0.042 sec)
```

```
MariaDB [EliteVideo]> create table price(Price_Code int primary key,  
Price_Description varchar(40) NOT NULL, Price_RentFee Decimal(6,2) NOT NULL,  
Price_DailyLateFee Decimal(6,2) NOT NULL);  
Query OK, 0 rows affected (0.025 sec)
```

```
MariaDB [EliteVideo]> create table movie( Movie_Num int primary key, Movie_Title  
varchar(90) NOT NULL, Movie_Year Year, Movie_Cost Decimal(8,2), Movie_Genre  
varchar(20), Price_Code int, constraint fk_movie_price foreign key(Price_Code)  
references price(Price_Code) on update cascade on delete set NULL);  
Query OK, 0 rows affected (0.030 sec)
```

```
MariaDB [EliteVideo]> create table video( Vid_Num int primary key, Vid_Indate DATE  
NOT NULL, Movie_Num int NOT NULL, constraint fk_video_movie foreign key(Movie_Num)  
references movie(Movie_Num) on update cascade on delete restrict);  
Query OK, 0 rows affected (0.028 sec)
```

```
MariaDB [EliteVideo]> create table rental(Rent_Num int primary key, Rent_Date DATE  
NOT NULL, Mem_Num int NOT NULL, constraint fk_rental_membership foreign  
key(Mem_Num) references MEMBERSHIP(Mem_Num) on update cascade on delete restrict);  
Query OK, 0 rows affected (0.041 sec)
```

```
MariaDB [EliteVideo]> create table detailrental(Rent_Num int, Vid_Num int,  
Detail_Fee Decimal(6,2) NOT NULL, Detail_DueDate DATE NOT NULL, Detail_ReturnDate  
DATE, Detail_DailyLateFee Decimal(6,2), primary key (Rent_Num, Vid_Num), constraint  
fk_detailrental_rental foreign key(Rent_Num) references rental(Rent_Num) on update  
cascade on delete cascade, constraint fk_detailrental_video foreign key(Vid_Num)  
references video(Vid_Num) on update cascade on delete restrict);  
Query OK, 0 rows affected (0.042 sec)
```

Question 2:

```
MariaDB [EliteVideo]> insert into MEMBERSHIP(Mem_Num, Mem_Fname, Mem_Lname,
Mem_Street, Mem_City, Mem_State, Mem_Zip, Mem_Balance) values (102, 'Tami',
'Dawson', '2632 Takli Circle', 'Norene', 'TN', '37136', 11.00), (103, 'Curt',
'Knight', '4025 Cornell Court', 'Flatgap', 'KY', '41219', 6.00), (104, 'Jamal',
'Melendez', '788 East 145th Avenue', 'Quebeck', 'TN', '38579', 0.00), (105, 'Iva',
'Mcclain', '6045 Musket Ball Circle', 'Summit', 'KY', '42783', 15.00), (106,
'Miranda', 'Parks', '4469 Maxwell Place', 'Germantown', 'TN', '38183', 0.00), (107,
'Rosario', 'Elliott', '7578 Danner Avenue', 'Columbia', 'TN', '38402', 5.00), (108,
'Mattie', 'Guy', '4390 Evergreen Street', 'Lily', 'KY', '40740', 0.00), (109,
'Clint', 'Ochoa', '1711 Elm Street', 'Greeneville', 'TN', '37745', 10.00), (110,
'Lewis', 'Rosales', '4524 Southwind Circle', 'Counce', 'TN', '38326', 0.00), (111,
'Stacy', 'Mann', '2789 East Cook Avenue', 'Murfreesboro', 'TN', '37132', 8.00),
(112, 'Luis', 'Trujillo', '7267 Melvin Avenue', 'Heiskell', 'TN', '37754', 3.00),
(113, 'Minnie', 'Gonzales', '6430 Vasili Drive', 'Williston', 'TN', '38076', 0.00);
Query OK, 12 rows affected (0.026 sec)
Records: 12 Duplicates: 0 Warnings: 0
```

```
MariaDB [EliteVideo]> insert into price (Price_Code, Price_Description,
Price_RentFee, Price_DailyLateFee) values (1, 'Standard', 2.00, 1.00), (2, 'New
Release', 3.50, 3.00), (3, 'Discount', 1.50, 1.00), (4, 'Weekly Special', 1.00,
0.50);
Query OK, 4 rows affected (0.016 sec)
Records: 4 Duplicates: 0 Warnings: 0
```

```
MariaDB [EliteVideo]> insert into movie (Movie_Num, Movie_Title, Movie_Year,
Movie_Cost, Movie_Genre, Price_Code) values (1234, 'The Cesar Family Christmas',
2014, 39.95, 'FAMILY', 2), (1235, 'Smokey Mountain Wildlife', 2011, 59.95,
'ACTION', 1), (1236, 'Richard Goodhope', 2015, 59.95, 'DRAMA', 2), (1237, 'Beatnik
Fever', 2014, 29.95, 'COMEDY', 2), (1238, 'Constant Companion', 2015, 89.95,
'DRAMA', NULL), (1239, 'Where Hope Dies', 2005, 25.49, 'DRAMA', 3), (1245, 'Time to
Burn', 2012, 45.49, 'ACTION', 1), (1246, 'What He Doesn''t Know', 2013, 58.29,
'COMEDY', 1);
Query OK, 8 rows affected (0.015 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

```
MariaDB [EliteVideo]> insert into video (Vid_Num, Vid_Indate, Movie_Num) values
(54321, '2015-06-18', 1234), (54324, '2015-06-18', 1234), (54325, '2015-06-18',
1234), (34341, '2014-01-22', 1235), (34342, '2014-01-22', 1235), (34366,
'2016-03-02', 1236), (34367, '2016-03-02', 1236), (34368, '2016-03-02', 1236),
(34369, '2016-03-02', 1236), (44392, '2015-10-21', 1237), (44397, '2015-10-21',
1237), (59237, '2016-02-14', 1237), (61388, '2014-01-25', 1239), (61353,
'2013-01-28', 1245), (61354, '2013-01-28', 1245), (61367, '2015-07-30', 1246),
(61369, '2015-07-30', 1246);
Query OK, 17 rows affected (0.016 sec)
Records: 17 Duplicates: 0 Warnings: 0
```

```
MariaDB [EliteVideo]> insert into rental(Rent_Num, Rent_Date, Mem_Num) values
(1001, '2016-03-01', 103), (1002, '2016-03-01', 105), (1003, '2016-03-02', 102),
```

```
(1004, '2016-03-02', 110), (1005, '2016-03-02', 111), (1006, '2016-03-02', 107),
(1007, '2016-03-02', 104), (1008, '2016-03-03', 105), (1009, '2016-03-03', 111);
Query OK, 9 rows affected (0.014 sec)
Records: 9 Duplicates: 0 Warnings: 0
```

```
MariaDB [EliteVideo]> insert into detailrental(Rent_Num, Vid_Num, Detail_Fee,
Detail_DueDate, Detail_ReturnDate, Detail_DailyLateFee) values (1001, 34342, 2.00,
'2016-03-04', '2016-03-02', NULL), (1001, 61353, 2.00, '2016-03-04', '2016-03-03',
1.00), (1002, 59237, 3.50, '2016-03-04', '2016-03-04', 3.00), (1003, 54325, 3.50,
'2016-03-04', '2016-03-09', 3.00), (1003, 61369, 2.00, '2016-03-06', '2016-03-09',
1.00), (1003, 61388, 0.00, '2016-03-06', '2016-03-09', 1.00), (1004, 44392, 3.50,
'2016-03-05', '2016-03-07', 3.00), (1004, 34367, 3.50, '2016-03-05', '2016-03-07',
3.00), (1004, 34341, 2.00, '2016-03-07', '2016-03-07', 1.00), (1005, 34342, 2.00,
'2016-03-07', '2016-03-05', 1.00), (1005, 44397, 3.50, '2016-03-05', '2016-03-05',
3.00), (1006, 34366, 3.50, '2016-03-05', '2016-03-04', 3.00), (1006, 61367, 2.00,
'2016-03-07', NULL, 1.00), (1007, 34368, 3.50, '2016-03-05', NULL, 3.00), (1008,
34369, 3.50, '2016-03-05', '2016-03-05', 3.00), (1009, 54324, 3.50, '2016-03-05',
NULL, 3.00), (1001, 34366, 3.50, '2016-03-04', '2016-03-02', 3.00);
Query OK, 17 rows affected (0.015 sec)
Records: 17 Duplicates: 0 Warnings: 0
```

Question 3:

```
MariaDB [EliteVideo]> select Movie_Num, Movie_Title, Movie_Year from movie where
Movie_Num = 1245;
```

Movie_Num	Movie_Title	Movie_Year
1245	Time to Burn	2012

1 row in set (0.021 sec)

```
MariaDB [EliteVideo]> update movie set Movie_Year = 2010 where Movie_Num = 1245;
Query OK, 1 row affected (0.017 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
MariaDB [EliteVideo]> select Movie_Num, Movie_Title, Movie_Year from movie where
Movie_Num = 1245;
```

Movie_Num	Movie_Title	Movie_Year
1245	Time to Burn	2010

1 row in set (0.001 sec)

Question 4:

```
MariaDB [EliteVideo]> select Movie_Num, Movie_Title, Price_Code from movie where
Movie_Genre = 'Action';
```

Movie_Num	Movie_Title	Price_Code
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Movie_Num	Movie_Title	Price_Code
1235	Smokey Mountain Wildlife	1
1245	Time to Burn	1

2 rows in set (0.015 sec)

MariaDB [EliteVideo]> update movie set Price_Code = 3 where Movie_Genre = 'Action';
 Query OK, 2 rows affected (0.005 sec)
 Rows matched: 2 Changed: 2 Warnings: 0

MariaDB [EliteVideo]> select Movie_Num, Movie_Title, Price_Code from movie where Movie_Genre = 'Action';

Movie_Num	Movie_Title	Price_Code
1235	Smokey Mountain Wildlife	3
1245	Time to Burn	3

2 rows in set (0.001 sec)

Question 5:

MariaDB [EliteVideo]> select Price_Code, Price_Description, Price_RentFee from price;

Price_Code	Price_Description	Price_RentFee
1	Standard	2.00
2	New Release	3.50
3	Discount	1.50
4	Weekly Special	1.00

4 rows in set (0.001 sec)

MariaDB [EliteVideo]> update price set Price_RentFee = Price_RentFee + 0.50;
 Query OK, 4 rows affected (0.016 sec)
 Rows matched: 4 Changed: 4 Warnings: 0

MariaDB [EliteVideo]> select Price_Code, Price_Description, Price_RentFee from price;

Price_Code	Price_Description	Price_RentFee
1	Standard	2.50
2	New Release	4.00
3	Discount	2.00
4	Weekly Special	1.50

4 rows in set (0.001 sec)

Question 6:

```
MariaDB [EliteVideo]> select Movie_Title, Movie_Year, Movie_Genre from movie;
```

Movie_Title	Movie_Year	Movie_Genre
The Cesar Family Christmas	2014	FAMILY
Smokey Mountain Wildlife	2011	ACTION
Richard Goodhope	2015	DRAMA
Beatnik Fever	2014	COMEDY
Constant Companion	2015	DRAMA
Where Hope Dies	2005	DRAMA
Time to Burn	2010	ACTION
What He Doesn't Know	2013	COMEDY

8 rows in set (0.001 sec)

Question 7:

```
MariaDB [EliteVideo]> select Movie_Year, Movie_Title, Movie_Cost from movie order by Movie_Year desc;
```

Movie_Year	Movie_Title	Movie_Cost
2015	Richard Goodhope	59.95
2015	Constant Companion	89.95
2014	The Cesar Family Christmas	39.95
2014	Beatnik Fever	29.95
2013	What He Doesn't Know	58.29
2011	Smokey Mountain Wildlife	59.95
2010	Time to Burn	45.49
2005	Where Hope Dies	25.49

8 rows in set (0.013 sec)

Question 8:

```
MariaDB [EliteVideo]> select Movie_Title, Movie_Year, Movie_Genre from movie order by Movie_Genre asc, Movie_Year desc;
```

Movie_Title	Movie_Year	Movie_Genre
Smokey Mountain Wildlife	2011	ACTION
Time to Burn	2010	ACTION
Beatnik Fever	2014	COMEDY
What He Doesn't Know	2013	COMEDY
Richard Goodhope	2015	DRAMA
Constant Companion	2015	DRAMA
Where Hope Dies	2005	DRAMA

The Cesar Family Christmas	2014	FAMILY
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8 rows in set (0.014 sec)

Question 9:

MariaDB [EliteVideo]> select Movie_Num, Movie_Title, Price_Code from movie where Movie_Title like 'R%';

Movie_Num	Movie_Title	Price_Code
1236	Richard Goodhope	2

1 row in set (0.012 sec)

Question 10:

MariaDB [EliteVideo]> select Movie_Title, Movie_Year, Movie_Cost from movie where Movie_Title like '%hope%' order by Movie_Title asc;

Movie_Title	Movie_Year	Movie_Cost
Richard Goodhope	2015	59.95
Where Hope Dies	2005	25.49

2 rows in set (0.010 sec)

Question 11:

MariaDB [EliteVideo]> select Movie_Title, Movie_Year, Movie_Genre from movie where Movie_Genre = 'Action';

Movie_Title	Movie_Year	Movie_Genre
Smokey Mountain Wildlife	2011	ACTION
Time to Burn	2010	ACTION

2 rows in set (0.001 sec)

Question 12:

MariaDB [EliteVideo]> select Movie_Num, Movie_Title, Movie_Cost from movie where Movie_Cost > 40;

Movie_Num	Movie_Title	Movie_Cost
1235	Smokey Mountain Wildlife	59.95
1236	Richard Goodhope	59.95
1238	Constant Companion	89.95
1245	Time to Burn	45.49

1246	What He Doesn't Know	58.29
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5 rows in set (0.011 sec)

Question 13:

MariaDB [EliteVideo]> select Movie_Num, Movie_Title, Movie_Cost, Movie_Genre from movie where Movie_Genre in('Action', 'Comedy') and Movie_Cost < 50 order by Movie_Genre asc;

Movie_Num	Movie_Title	Movie_Cost	Movie_Genre
1245	Time to Burn	45.49	ACTION
1237	Beatnik Fever	29.95	COMEDY

2 rows in set (0.007 sec)

Question 14:

MariaDB [EliteVideo]> select Mem_Num, Mem_Fname, Mem_Lname, Mem_Street, Mem_Balance from MEMBERSHIP where Mem_State = 'TN' and Mem_Balance < 5 and Mem_Street like '%Avenue';

Mem_Num	Mem_Fname	Mem_Lname	Mem_Street	Mem_Balance
104	Jamal	Melendez	788 East 145th Avenue	0.00
112	Luis	Trujillo	7267 Melvin Avenue	3.00

2 rows in set (0.015 sec)

Question 15:

MariaDB [EliteVideo]> select Movie_Genre, COUNT(*) AS Number_Of_Movies from movie group by Movie_Genre;

Movie_Genre	Number_Of_Movies
ACTION	2
COMEDY	2
DRAMA	3
FAMILY	1

4 rows in set (0.001 sec)

Question 16:

MariaDB [EliteVideo]> select avg(Movie_Cost) as Average_Movie_Cost from movie;

Average_Movie_Cost

```
|          51.127500 |
+-----+
1 row in set (0.013 sec)
```

Question 17:

```
MariaDB [EliteVideo]> select Movie_Genre, avg(Movie_Cost) as
Average_Cost_of_Movies from movie group by Movie_Genre;
```

```
+-----+-----+
| Movie_Genre | Average_Cost_of_Movies |
+-----+-----+
| ACTION      | 52.720000              |
| COMEDY      | 44.120000              |
| DRAMA       | 58.463333              |
| FAMILY      | 39.950000              |
+-----+-----+
4 rows in set (0.013 sec)
```

Question 18:

```
MariaDB [EliteVideo]> select movie.Movie_Title, movie.Movie_Genre,
price.Price_Description, price.Price_RentFee from movie join price on
movie.Price_Code = price.Price_Code;
```

```
+-----+-----+-----+-----+
| Movie_Title          | Movie_Genre | Price_Description | Price_RentFee |
+-----+-----+-----+-----+
| The Cesar Family Christmas | FAMILY      | New Release      | 4.00          |
| Smokey Mountain Wildlife  | ACTION      | Discount         | 2.00          |
| Richard Goodhope         | DRAMA       | New Release      | 4.00          |
| Beatnik Fever            | COMEDY      | New Release      | 4.00          |
| Where Hope Dies          | DRAMA       | Discount         | 2.00          |
| Time to Burn             | ACTION      | Discount         | 2.00          |
| What He Doesn't Know     | COMEDY      | Standard         | 2.50          |
+-----+-----+-----+-----+
7 rows in set (0.015 sec)
```

Question 19:

```
MariaDB [EliteVideo]> select movie.Movie_Genre, ROUND(avg(price.Price_RentFee), 2)
as Average_Rental_Fee from movie join price on movie.Price_Code = price.Price_Code
group by movie.Movie_Genre;
```

```
+-----+-----+
| Movie_Genre | Average_Rental_Fee |
+-----+-----+
| ACTION      | 2.00               |
| COMEDY      | 3.25               |
| DRAMA       | 3.00               |
| FAMILY      | 4.00               |
+-----+-----+
```



```
+-----+-----+
4 rows in set (0.053 sec)
```

Question 20:

```
MariaDB [EliteVideo]> select movie.Movie_Title, Round(movie.Movie_Cost /
price.Price_RentFee, 2) as BreakEven_Rentals from movie join price on
movie.Price_Code = price.Price_Code;
```

Movie_Title	BreakEven_Rentals
The Cesar Family Christmas	9.99
Smokey Mountain Wildlife	29.98
Richard Goodhope	14.99
Beatnik Fever	7.49
Where Hope Dies	12.75
Time to Burn	22.75
What He Doesn't Know	23.32

```
7 rows in set (0.016 sec)
```

Question 21:

```
MariaDB [EliteVideo]> select Movie_Title, Movie_Year from movie where Price_Code IS
NOT NULL;
```

Movie_Title	Movie_Year
What He Doesn't Know	2013
The Cesar Family Christmas	2014
Richard Goodhope	2015
Beatnik Fever	2014
Smokey Mountain Wildlife	2011
Where Hope Dies	2005
Time to Burn	2010

```
7 rows in set (0.018 sec)
```

Question 22:

```
MariaDB [EliteVideo]> select Movie_Title, Movie_Genre, Movie_Cost from movie where
Movie_Cost between 44.99 and 49.99;
```

Movie_Title	Movie_Genre	Movie_Cost
Time to Burn	ACTION	45.49

```
1 row in set (0.004 sec)
```

Question 23:

```
MariaDB [EliteVideo]> select movie.Movie_Title, price.Price_Description,  
price.Price_RentFee from movie join price on movie.Price_Code = price.Price_Code  
where Movie_Genre IN ('Family','Comedy','Drama');
```

Movie_Title	Price_Description	Price_RentFee
The Cesar Family Christmas	New Release	4.00
Richard Goodhope	New Release	4.00
Beatnik Fever	New Release	4.00
Where Hope Dies	Discount	2.00
What He Doesn't Know	Standard	2.50

5 rows in set (0.015 sec)

Question 24:

```
MariaDB [EliteVideo]> select distinct MEMBERSHIP.Mem_Num, MEMBERSHIP.Mem_Fname,  
MEMBERSHIP.Mem_Lname, MEMBERSHIP.Mem_Balance from MEMBERSHIP join rental on  
MEMBERSHIP.Mem_Num = rental.Mem_Num;
```

Mem_Num	Mem_Fname	Mem_Lname	Mem_Balance
102	Tami	Dawson	11.00
103	Curt	Knight	6.00
104	Jamal	Melendez	0.00
105	Iva	Mcclain	15.00
107	Rosario	Elliott	5.00
110	Lewis	Rosales	0.00
111	Stacy	Mann	8.00

7 rows in set (0.015 sec)

Question 25:

```
MariaDB [EliteVideo]> select min(Mem_Balance) as Minimum_Balance, max(Mem_Balance)  
as Maximum_Balance, Round(avg(Mem_Balance),2) as Average_Balance from ( select  
distinct MEMBERSHIP.Mem_Num, MEMBERSHIP.Mem_Balance from MEMBERSHIP join rental on  
MEMBERSHIP.Mem_Num = rental.Mem_Num) as Members_With_Rentals;
```

Minimum_Balance	Maximum_Balance	Average_Balance
0.00	15.00	6.43

1 row in set (0.018 sec)

Question 26:

```
MariaDB [EliteVideo]> select rental.Rent_Num, rental.Rent_Date,
detailrental.Vid_Num, movie.Movie_Title, detailrental.Detail_DueDate,
detailrental.Detail_ReturnDate from rental join detailrental on rental.Rent_Num =
detailrental.Rent_Num join video on detailrental.Vid_Num = video.Vid_Num join movie
on video.Movie_Num = movie.Movie_Num where detailrental.Detail_ReturnDate >
detailrental.Detail_DueDate order by rental.Rent_Num asc, movie.Movie_Title asc;
```

Rent_Num	Rent_Date	Vid_Num	Movie_Title	Detail_DueDate	Detail_ReturnDate
1003	2016-03-02	54325	The Cesar Family Christmas	2016-03-04	2016-03-09
1003	2016-03-02	61369	What He Doesn't Know	2016-03-06	2016-03-09
1003	2016-03-02	61388	Where Hope Dies	2016-03-06	2016-03-09
1004	2016-03-02	44392	Beatnik Fever	2016-03-05	2016-03-07
1004	2016-03-02	34367	Richard Goodhope	2016-03-05	2016-03-07

5 rows in set (0.021 sec)

Question 27:

```
MariaDB [EliteVideo]> select rental.Rent_Num, rental.Rent_Date, movie.Movie_Title,
detailrental.Detail_Fee from rental join detailrental on rental.Rent_Num =
detailrental.Rent_Num join video on detailrental.Vid_Num = video.Vid_Num join movie
on video.Movie_Num = movie.Movie_Num where detailrental.Detail_ReturnDate <=
detailrental.Detail_DueDate;
```

Rent_Num	Rent_Date	Movie_Title	Detail_Fee
1001	2016-03-01	Smokey Mountain Wildlife	2.00
1001	2016-03-01	Richard Goodhope	3.50
1001	2016-03-01	Time to Burn	2.00
1002	2016-03-01	Beatnik Fever	3.50
1004	2016-03-02	Smokey Mountain Wildlife	2.00
1005	2016-03-02	Smokey Mountain Wildlife	2.00
1005	2016-03-02	Beatnik Fever	3.50
1006	2016-03-02	Richard Goodhope	3.50
1008	2016-03-03	Richard Goodhope	3.50

Question 28:

```
MariaDB [EliteVideo]> select movie.Movie_Num, movie.Movie_Genre,
ROUND(g.Average_Genre_Cost, 2) as Average_Genre_Cost, movie.Movie_Cost, ROUND
(((movie.Movie_Cost - g.Average_Genre_Cost)/ g.Average_Genre_Cost) * 100, 2 ) as
Percentage_Difference from movie join ( select Movie_Genre, avg(Movie_Cost) as
Average_Genre_Cost from movie group by Movie_Genre) as g on movie.Movie_Genre =
g.Movie_Genre order by movie.Movie_Genre, movie.Movie_Num;
```

Movie_Num	Movie_Genre	Average_Genre_Cost	Movie_Cost	Percentage_Difference
1235	ACTION	52.72	59.95	13.71
1245	ACTION	52.72	45.49	-13.71
1237	COMEDY	44.12	29.95	-32.12
1246	COMEDY	44.12	58.29	32.12
1236	DRAMA	58.46	59.95	2.54
1238	DRAMA	58.46	89.95	53.86
1239	DRAMA	58.46	25.49	-56.40
1234	FAMILY	39.95	39.95	0.00

```
8 rows in set (0.022 sec)
```